GoFixCarz Admin API — frontend handoff

Share this document and `GoFixCarz_Admin_API_OpenAPI.json` (machine-readable OpenAPI 3.1). The Postman collection is `GoFixCarz_Admin_APIs.postman_collection.json`.

API host and exact URL format

A stable public/production hostname has not been confirmed for this API, so this handoff does not invent one. Configure `API_BASE_URL` to the API origin reachable from the frontend (protocol + host + optional port, no trailing slash).

- Local development origin: `http://localhost:8080` (only if the backend is running on the same machine).
- Protected endpoint URLs: `{{API_BASE_URL}}/api/v1/admin/...`.
- Private auth endpoint URLs: `{{API_BASE_URL}}/api/v1/admin-auth/...`.
- Concrete local example: `http://localhost:8080/api/v1/admin/me`.
- Swagger: `{{API_BASE_URL}}/api/docs`; OpenAPI: `{{API_BASE_URL}}/api/openapi.json`.

Replace `API_BASE_URL` with the dev/staging host that the frontend can reach. The OpenAPI file includes a local server and a replaceable host variable.

Authentication

1. `POST /api/v1/admin-auth/sign-in` with administrator email (or the accepted `identifier` alias) and password. If `mfa_required` is true, copy `challenge_id` and call `/api/v1/admin-auth/mfa` with the current six-digit TOTP. On first enrollment, the password-verified sign-in response also contains a secret `provisioning_uri` for an authenticator.

2. Save `access_token` and `refresh_token` in secure client storage. Send `Authorization: Bearer <access_token>` for every `/api/v1/admin/*` request and `/admin-auth/logout`.

3. Access tokens expire after 15 minutes. Refresh rotates the refresh token; replace both tokens after every successful refresh. TOTP challenges expire after five minutes and are single-use.

4. Use `/me` and `/menus` to build navigation. The backend permission-checks every request; a hidden frontend button is not authorization.

Response format

JSON success envelope:

```
{
  "success": true,
  "message": "Success",
  "data": {},
  "errors": [],
  "status_code": 200
}
```

Endpoint entries below describe `data`. Validation errors use the standard envelope with `errors: [{field, message}]`; admin auth failures also use the envelope. Some business rules raise FastAPI HTTP errors in `{"detail": ...}` format. Typical statuses: 400 (invalid MFA proof), 401 (auth), 403 (permission/scope), 404 (missing/out-of-scope resource), 409 (version/idempotency/state conflict), 422 (validation), and 429 (rate limit). Export download returns CSV/XLSX bytes, not JSON.

Endpoint URLs, request parameters, payloads and responses

Total: 70 operations (66 protected admin operations and 4 private auth operations). Local URL templates use `http://localhost:8080`; replace that origin for a remote environment.

POST `{{API_BASE_URL}}/api/v1/admin-auth/sign-in`

Local test URL: `http://localhost:8080/api/v1/admin-auth/sign-in`

Authentication: Private auth request; no bearer token

Summary: Admin sign-in (private)

Request JSON:

```
{
  "email": "{{adminEmail}}",
  "password": "{{adminPassword}}",
  "device_name": "Postman development test"
}
```

Success: HTTP 200. `data` = MFA challenge or token pair. Challenge: challenge_id, mfa_required, mfa_method, expires_in_seconds, enrollment_required, optional provisioning_uri. Token response: access_token, refresh_token, token_type, expires_in, user { id, email, name, role, is_active, role_ids, permissions, data_scope }.

POST `{{API_BASE_URL}}/api/v1/admin-auth/mfa`

Local test URL: `http://localhost:8080/api/v1/admin-auth/mfa`

Authentication: Private auth request; no bearer token

Summary: Complete admin MFA (private)

Request JSON:

```
{
  "challenge_id": "{{mfaChallengeId}}",
  "code": "123456"
}
```

Success: HTTP 200. `data` = Token response: access_token, refresh_token, token_type, expires_in, user { id, email, name, role, is_active, role_ids, permissions, data_scope }.

POST `{{API_BASE_URL}}/api/v1/admin-auth/refresh`

Local test URL: `http://localhost:8080/api/v1/admin-auth/refresh`

Authentication: Private auth request; no bearer token

Summary: Refresh admin session (private)

Request JSON:

```
{
  "refresh_token": "{{refreshToken}}"
}
```

Success: HTTP 200. `data` = Token response: access_token, refresh_token, token_type, expires_in, user { id, email, name, role, is_active, role_ids, permissions, data_scope }.

POST `{{API_BASE_URL}}/api/v1/admin-auth/logout`

Local test URL: `http://localhost:8080/api/v1/admin-auth/logout`

Authentication: Bearer admin access token

Summary: Admin logout (private)

Success: HTTP 200. `data` = { logged_out: true }

Admin

GET `{{API_BASE_URL}}/api/v1/admin/analytics`

Local test URL: `http://localhost:8080/api/v1/admin/analytics`

Authentication: Bearer admin access token

Summary: Analytics

Parameters:

- `period` — query; optional; string; allowed values: daily, weekly, monthly, custom; default: monthly
- `start_date` — query; optional
- `end_date` — query; optional
- `snapshot_id` — query; optional
- `date_from` — query; optional
- `date_to` — query; optional
- `timezone` — query; optional; string; default: UTC
- `city` — query; optional
- `garage_id` — query; optional

Success: HTTP 200. `data` = Analytics summary: period, start_date, end_date, total_jobs, total_revenue, status_counts, graph_data, snapshot_id, generated_at.

GET `{{API_BASE_URL}}/api/v1/admin/audit-logs`

Local test URL: `http://localhost:8080/api/v1/admin/audit-logs`

Authentication: Bearer admin access token

Summary: Audit Logs

Parameters:

- `actor_id` — query; optional
- `action` — query; optional
- `date_from` — query; optional
- `date_to` — query; optional
- `timezone` — query; optional; string; default: UTC
- `page` — query; optional; integer; default: 1
- `page_size` — query; optional; integer; default: 20

Success: HTTP 200. `data` = Response data schema: components.schemas.PaginatedData_AdminAuditResponse_.

GET `{{API_BASE_URL}}/api/v1/admin/billing/issues`

Local test URL: `http://localhost:8080/api/v1/admin/billing/issues`

Authentication: Bearer admin access token

Summary: Billing Issues

Parameters:

- `garage_id` — query; optional
- `page` — query; optional; integer; default: 1
- `page_size` — query; optional; integer; default: 20

Success: HTTP 200. `data` = Response data schema: components.schemas.PaginatedData_JobResponse_.

GET `{{API_BASE_URL}}/api/v1/admin/bookings`

Local test URL: `http://localhost:8080/api/v1/admin/bookings`

Authentication: Bearer admin access token

Summary: Bookings

Parameters:

- `customer_id` — query; optional
- `search` — query; optional
- `status` — query; optional
- `page` — query; optional; integer; default: 1
- `page_size` — query; optional; integer; default: 20
- `date_from` — query; optional
- `date_to` — query; optional
- `timezone` — query; optional; string; default: UTC
- `city` — query; optional
- `garage_id` — query; optional

Success: HTTP 200. `data` = Response data schema: components.schemas.PaginatedData_BookingResponse_.

GET `{{API_BASE_URL}}/api/v1/admin/dashboard`

Local test URL: `http://localhost:8080/api/v1/admin/dashboard`

Authentication: Bearer admin access token

Summary: Dashboard

Parameters:

- `snapshot_id` — query; optional
- `date_from` — query; optional
- `date_to` — query; optional
- `timezone` — query; optional; string; default: UTC
- `city` — query; optional
- `garage_id` — query; optional

Success: HTTP 200. `data` = Dashboard counts/status_counts and revenue, plus snapshot_id, generated_at, extended_metrics and metric definitions.

GET `{{API_BASE_URL}}/api/v1/admin/garages`

Local test URL: `http://localhost:8080/api/v1/admin/garages`

Authentication: Bearer admin access token

Summary: Garages

Parameters:

- `search` — query; optional
- `user_id` — query; optional

- `city` — query; optional
- `operational_status` — query; optional
- `verification_status` — query; optional
- `page` — query; optional; integer; default: 1
- `page_size` — query; optional; integer; default: 20

Success: HTTP 200. `data` = Response data schema: components.schemas.PaginatedData_GarageResponse_.

GET `{{API_BASE_URL}}/api/v1/admin/garages/{garage_id}`

Local test URL: `http://localhost:8080/api/v1/admin/garages/<garage_id>`

Authentication: Bearer admin access token

Summary: Garage Detail

Parameters:

- `garage_id` — path; required; string

Success: HTTP 200. `data` = Response data schema: components.schemas.AdminGarageDetail.

GET `{{API_BASE_URL}}/api/v1/admin/jobs`

Local test URL: `http://localhost:8080/api/v1/admin/jobs`

Authentication: Bearer admin access token

Summary: Jobs

Parameters:

- `status` — query; optional
- `search` — query; optional
- `customer_id` — query; optional
- `booking_id` — query; optional
- `page` — query; optional; integer; default: 1
- `page_size` — query; optional; integer; default: 20
- `date_from` — query; optional
- `date_to` — query; optional
- `timezone` — query; optional; string; default: UTC
- `city` — query; optional
- `garage_id` — query; optional

Success: HTTP 200. `data` = Response data schema: components.schemas.PaginatedData_JobResponse_.

GET `{{API_BASE_URL}}/api/v1/admin/jobs/{job_id}`

Local test URL: `http://localhost:8080/api/v1/admin/jobs/<job_id>`

Authentication: Bearer admin access token

Summary: Job Detail

Parameters:

- `job_id` — path; required; string

Success: HTTP 200. `data` = Response data schema: components.schemas.JobDetailResponse.

GET `{{API_BASE_URL}}/api/v1/admin/me`

Local test URL: `http://localhost:8080/api/v1/admin/me`

Authentication: Bearer admin access token

Summary: Admin Me

Success: HTTP 200. `data` = Admin identity: id, email, name, role, is_active, permissions, role_ids, data_scope.

GET `{{API_BASE_URL}}/api/v1/admin/menus`

Local test URL: `http://localhost:8080/api/v1/admin/menus`

Authentication: Bearer admin access token

Summary: Admin Menus

Success: HTTP 200. `data` = { menus: [{ key, label, api }] }

GET `{{API_BASE_URL}}/api/v1/admin/users`

Local test URL: `http://localhost:8080/api/v1/admin/users`

Authentication: Bearer admin access token

Summary: Users

Parameters:

- `search` — query; optional
- `is_active` — query; optional
- `role` — query; optional
- `include_deleted` — query; optional; boolean; default: false
- `page` — query; optional; integer; default: 1
- `page_size` — query; optional; integer; default: 20

Success: HTTP 200. `data` = Response data schema: components.schemas.PaginatedData_UserResponse_.

GET `{{API_BASE_URL}}/api/v1/admin/users/{user_id}`

Local test URL: `http://localhost:8080/api/v1/admin/users/<user_id>`

Authentication: Bearer admin access token

Summary: User Detail

Parameters:

- `user_id` — path; required; string
- `include_deleted` — query; optional; boolean; default: false

Success: HTTP 200. `data` = Response data schema: components.schemas.AdminUserDetail.

PATCH `{{API_BASE_URL}}/api/v1/admin/users/{user_id}/status`

Local test URL: `http://localhost:8080/api/v1/admin/users/<user_id>/status`

Authentication: Bearer admin access token

Summary: User Status

Parameters:

- `user_id` — path; required; string

Request JSON:

```
{
  "is_active": true
}
```

Success: HTTP 200. `data` = Response data schema: components.schemas.UserResponse.

Admin Finance

GET `{{API_BASE_URL}}/api/v1/admin/disputes`

Local test URL: `http://localhost:8080/api/v1/admin/disputes`

Authentication: Bearer admin access token

Summary: Disputes

Parameters:

- `status` — query; optional
- `customer_id` — query; optional
- `page` — query; optional; integer; default: 1
- `page_size` — query; optional; integer; default: 20
- `date_from` — query; optional
- `date_to` — query; optional
- `timezone` — query; optional; string; default: UTC
- `city` — query; optional
- `garage_id` — query; optional

Success: HTTP 200. `data` = SourceRecord detail or paginated source records. Lists include items/total/page/page_size/pages, filters, money_unit: paise, currency: INR, source_status.

GET `{{API_BASE_URL}}/api/v1/admin/disputes/{dispute_id}`

Local test URL: `http://localhost:8080/api/v1/admin/disputes/<dispute_id>`

Authentication: Bearer admin access token

Summary: Dispute Detail

Parameters:

- `dispute_id` — path; required; string

Success: HTTP 200. `data` = SourceRecord detail or paginated source records. Lists include items/total/page/page_size/pages, filters, money_unit: paise, currency: INR, source_status.

GET `{{API_BASE_URL}}/api/v1/admin/finance/summary`

Local test URL: `http://localhost:8080/api/v1/admin/finance/summary`

Authentication: Bearer admin access token

Summary: Finance Summary

Parameters:

- `snapshot_id` — query; optional
- `date_from` — query; optional
- `date_to` — query; optional
- `timezone` — query; optional; string; default: UTC
- `city` — query; optional
- `garage_id` — query; optional

Success: HTTP 200. `data` = Finance snapshot metadata and totals in integer INR paise: gross_collected_amount, refunded_amount, net_collected_amount, captured_payments, failed_attempts, paid/issued invoice counts, invoice totals/average, source_status.

GET `{{API_BASE_URL}}/api/v1/admin/finance/trends`

Local test URL: `http://localhost:8080/api/v1/admin/finance/trends`

Authentication: Bearer admin access token

Summary: Finance Trends

Parameters:

- `snapshot_id` — query; optional
- `period` — query; optional; string; allowed values: daily, weekly, monthly; default: daily
- `date_from` — query; optional
- `date_to` — query; optional
- `timezone` — query; optional; string; default: UTC
- `city` — query; optional
- `garage_id` — query; optional

Success: HTTP 200. `data` = Finance snapshot metadata, period, series[] with date and gross/refund/net integer INR paise.

GET `{{API_BASE_URL}}/api/v1/admin/invoices`

Local test URL: `http://localhost:8080/api/v1/admin/invoices`

Authentication: Bearer admin access token

Summary: Invoices

Parameters:

- `status` — query; optional
- `customer_id` — query; optional
- `page` — query; optional; integer; default: 1
- `page_size` — query; optional; integer; default: 20
- `date_from` — query; optional
- `date_to` — query; optional
- `timezone` — query; optional; string; default: UTC
- `city` — query; optional
- `garage_id` — query; optional

Success: HTTP 200. `data` = SourceRecord detail or paginated source records. Lists include items/total/page/page_size/pages, filters, money_unit: paise, currency: INR, source_status.

GET `{{API_BASE_URL}}/api/v1/admin/invoices/{invoice_id}`

Local test URL: `http://localhost:8080/api/v1/admin/invoices/<invoice_id>`

Authentication: Bearer admin access token

Summary: Invoice Detail

Parameters:

- `invoice_id` — path; required; string

Success: HTTP 200. `data` = SourceRecord detail or paginated source records. Lists include items/total/page/page_size/pages, filters, money_unit: paise, currency: INR, source_status.

GET `{{API_BASE_URL}}/api/v1/admin/invoices/{invoice_id}/payment-attempts`

Local test URL: `http://localhost:8080/api/v1/admin/invoices/<invoice_id>/payment-attempts`

Authentication: Bearer admin access token

Summary: Payment Attempts

Parameters:

- `invoice_id` — path; required; string
- `page` — query; optional; integer; default: 1
- `page_size` — query; optional; integer; default: 20

Success: HTTP 200. `data` = SourceRecord detail or paginated source records. Lists include items/total/page/page_size/pages, filters, money_unit: paise, currency: INR, source_status.

GET `{{API_BASE_URL}}/api/v1/admin/payments`

Local test URL: `http://localhost:8080/api/v1/admin/payments`

Authentication: Bearer admin access token

Summary: Payments

Parameters:

- `status` — query; optional
- `customer_id` — query; optional
- `invoice_id` — query; optional
- `page` — query; optional; integer; default: 1
- `page_size` — query; optional; integer; default: 20
- `date_from` — query; optional
- `date_to` — query; optional
- `timezone` — query; optional; string; default: UTC
- `city` — query; optional
- `garage_id` — query; optional

Success: HTTP 200. `data` = SourceRecord detail or paginated source records. Lists include items/total/page/page_size/pages, filters, money_unit: paise, currency: INR, source_status.

GET `{{API_BASE_URL}}/api/v1/admin/payments/{payment_id}`

Local test URL: `http://localhost:8080/api/v1/admin/payments/<payment_id>`

Authentication: Bearer admin access token

Summary: Payment Detail

Parameters:

- `payment_id` — path; required; string

Success: HTTP 200. `data` = SourceRecord detail or paginated source records. Lists include items/total/page/page_size/pages, filters, money_unit: paise, currency: INR, source_status.

GET `{{API_BASE_URL}}/api/v1/admin/refunds`

Local test URL: `http://localhost:8080/api/v1/admin/refunds`

Authentication: Bearer admin access token

Summary: Refunds

Parameters:

- `status` — query; optional
- `customer_id` — query; optional
- `payment_id` — query; optional
- `page` — query; optional; integer; default: 1
- `page_size` — query; optional; integer; default: 20
- `date_from` — query; optional
- `date_to` — query; optional
- `timezone` — query; optional; string; default: UTC
- `city` — query; optional
- `garage_id` — query; optional

Success: HTTP 200. `data` = SourceRecord detail or paginated source records. Lists include items/total/page/page_size/pages, filters, money_unit: paise, currency: INR, source_status.

Admin Monitoring

GET `{{API_BASE_URL}}/api/v1/admin/activity`

Local test URL: `http://localhost:8080/api/v1/admin/activity`

Authentication: Bearer admin access token

Summary: Activity

Parameters:

- `owner_id` — query; optional
- `customer_id` — query; optional
- `booking_id` — query; optional
- `job_id` — query; optional
- `event_type` — query; optional

- `page` — query; optional; integer; default: 1
- `page_size` — query; optional; integer; default: 20
- `date_from` — query; optional
- `date_to` — query; optional
- `timezone` — query; optional; string; default: UTC
- `city` — query; optional
- `garage_id` — query; optional

Success: HTTP 200. `data` = Response data schema: components.schemas.AdminActivityPage.

GET `{{API_BASE_URL}}/api/v1/admin/activity/service-funnels/{booking_id}`

Local test URL: `http://localhost:8080/api/v1/admin/activity/service-funnels/<booking_id>`

Authentication: Bearer admin access token

Summary: Funnel

Parameters:

- `booking_id` — path; required; string
- `page` — query; optional; integer; default: 1
- `page_size` — query; optional; integer; default: 20

Success: HTTP 200. `data` = Response data schema: components.schemas.AdminFunnelResponse.

GET `{{API_BASE_URL}}/api/v1/admin/audit-logs/{audit_id}`

Local test URL: `http://localhost:8080/api/v1/admin/audit-logs/<audit_id>`

Authentication: Bearer admin access token

Summary: Audit Detail

Parameters:

- `audit_id` — path; required; string

Success: HTTP 200. `data` = Response data schema: components.schemas.AdminAuditResponse.

GET `{{API_BASE_URL}}/api/v1/admin/bookings/exception-statistics`

Local test URL: `http://localhost:8080/api/v1/admin/bookings/exception-statistics`

Authentication: Bearer admin access token

Summary: Booking Exceptions

Parameters:

- `group_by` — query; optional
- `date_from` — query; optional
- `date_to` — query; optional
- `timezone` — query; optional; string; default: UTC
- `city` — query; optional
- `garage_id` — query; optional

Success: HTTP 200. `data` = Exception aggregate: filters, generated_at, cohort, rejected/cancelled/expired counts, reason_counts, breakdown, availability metadata.

GET `{{API_BASE_URL}}/api/v1/admin/bookings/{booking_id}`

Local test URL: `http://localhost:8080/api/v1/admin/bookings/<booking_id>`

Authentication: Bearer admin access token

Summary: Booking Detail

Parameters:

- `booking_id` — path; required; string

Success: HTTP 200. `data` = Response data schema: components.schemas.AdminBookingDetail.

GET `{{API_BASE_URL}}/api/v1/admin/garage-owners`

Local test URL: `http://localhost:8080/api/v1/admin/garage-owners`

Authentication: Bearer admin access token

Summary: Owners

Parameters:

- `search` — query; optional
- `status` — query; optional
- `include_deleted` — query; optional; boolean; default: false
- `min_rating` — query; optional
- `page` — query; optional; integer; default: 1
- `page_size` — query; optional; integer; default: 20
- `date_from` — query; optional
- `date_to` — query; optional
- `timezone` — query; optional; string; default: UTC
- `city` — query; optional
- `garage_id` — query; optional

Success: HTTP 200. `data` = Response data schema: components.schemas.AdminOwnerPage.

GET `{{API_BASE_URL}}/api/v1/admin/garage-owners/{owner_id}`

Local test URL: `http://localhost:8080/api/v1/admin/garage-owners/<owner_id>`

Authentication: Bearer admin access token

Summary: Owner Detail

Parameters:

- `owner_id` — path; required; string
- `include_deleted` — query; optional; boolean; default: false
- `date_from` — query; optional
- `date_to` — query; optional
- `timezone` — query; optional; string; default: UTC
- `city` — query; optional

- `garage_id` — query; optional

Success: HTTP 200. `data` = Response data schema: components.schemas.AdminOwnerDetailResponse.

GET `{{API_BASE_URL}}/api/v1/admin/jobs/{job_id}/timeline`

Local test URL: `http://localhost:8080/api/v1/admin/jobs/<job_id>/timeline`

Authentication: Bearer admin access token

Summary: Timeline

Parameters:

- `job_id` — path; required; string
- `page` — query; optional; integer; default: 1
- `page_size` — query; optional; integer; default: 20

Success: HTTP 200. `data` = Response data schema: components.schemas.PaginatedData_AdminTimelineEvent_.

Admin Operations

GET `{{API_BASE_URL}}/api/v1/admin/alerts`

Local test URL: `http://localhost:8080/api/v1/admin/alerts`

Authentication: Bearer admin access token

Summary: Alerts

Parameters:

- `type` — query; optional; string; default: JOB_DELAY
- `severity` — query; optional
- `page` — query; optional; integer; default: 1
- `page_size` — query; optional; integer; default: 20
- `date_from` — query; optional
- `date_to` — query; optional
- `timezone` — query; optional; string; default: UTC
- `city` — query; optional
- `garage_id` — query; optional

Success: HTTP 200. `data` = { items: delay alerts (id, type, severity, job/state-age fields), total, page, page_size, as_of, generation }

GET `{{API_BASE_URL}}/api/v1/admin/announcements`

Local test URL: `http://localhost:8080/api/v1/admin/announcements`

Authentication: Bearer admin access token

Summary: Announcements

Parameters:

- `page` — query; optional; integer; default: 1
- `page_size` — query; optional; integer; default: 20

Success: HTTP 200. `data` = Announcement: id, status, title, message, delivery_count, created_at (send returns HTTP 202).

POST `{{API_BASE_URL}}/api/v1/admin/announcements`

Local test URL: `http://localhost:8080/api/v1/admin/announcements`

Authentication: Bearer admin access token

Summary: Send Announcement

Parameters:

- `Idempotency-Key` — header; required; string

Request JSON:

```
{
  "preview_id": "{{preview_id}}",
  "reason": "Postman development test"
}
```

Success: HTTP 202. `data` = Announcement: id, status, title, message, delivery_count, created_at (send returns HTTP 202).

POST `{{API_BASE_URL}}/api/v1/admin/announcements/preview`

Local test URL: `http://localhost:8080/api/v1/admin/announcements/preview`

Authentication: Bearer admin access token

Summary: Preview Announcement

Request JSON:

```
{
  "audience": {
    "recipient_roles": [
      "GARAGE_OWNER"
    ],
    "garage_ids": []
  },
  "channels": [
    "IN_APP"
  ],
  "title": "Development test announcement",
  "body": "Test message. Send only to development test accounts."
}
```

Success: HTTP 200. `data` = { preview_id, recipient_count, delivery_count, channels, expires_at }

GET `{{API_BASE_URL}}/api/v1/admin/announcements/{announcement_id}`

Local test URL: `http://localhost:8080/api/v1/admin/announcements/<announcement_id>`

Authentication: Bearer admin access token

Summary: Announcement Detail

Parameters:

- `announcement_id` — path; required; string

Success: HTTP 200. `data` = Announcement/delivery records; lists use items/total/page/page_size.

GET `{{API_BASE_URL}}/api/v1/admin/announcements/{announcement_id}/deliveries`

Local test URL: `http://localhost:8080/api/v1/admin/announcements/<announcement_id>/deliveries`

Authentication: Bearer admin access token

Summary: Announcement Deliveries

Parameters:

- `announcement_id` — path; required; string
- `status` — query; optional
- `page` — query; optional; integer; default: 1
- `page_size` — query; optional; integer; default: 20

Success: HTTP 200. `data` = Announcement/delivery records; lists use items/total/page/page_size.

GET `{{API_BASE_URL}}/api/v1/admin/customers`

Local test URL: `http://localhost:8080/api/v1/admin/customers`

Authentication: Bearer admin access token

Summary: Customers

Parameters:

- `page` — query; optional; integer; default: 1
- `page_size` — query; optional; integer; default: 20
- `account_status` — query; optional
- `has_failed_payments` — query; optional
- `has_support_flags` — query; optional
- `inactive_since` — query; optional
- `search` — query; optional

Success: HTTP 200. `data` = Customer list: items, total, page, page_size. Rows include id/name/contact/account_status/last_activity_at/garage_ids/support and payment-risk flags/source/restricted_fields.

GET `{{API_BASE_URL}}/api/v1/admin/customers/{customer_id}`

Local test URL: `http://localhost:8080/api/v1/admin/customers/<customer_id>`

Authentication: Bearer admin access token

Summary: Customer Detail

Parameters:

- `customer_id` — path; required; string

Success: HTTP 200. `data` = Customer account object with vehicles[], reviews[], preview_limit.

GET `{{API_BASE_URL}}/api/v1/admin/customers/{customer_id}/reviews`

Local test URL: `http://localhost:8080/api/v1/admin/customers/<customer_id>/reviews`

Authentication: Bearer admin access token

Summary: Customer Reviews

Parameters:

- `customer_id` — path; required; string
- `page` — query; optional; integer; default: 1
- `page_size` — query; optional; integer; default: 20

Success: HTTP 200. `data` = Paginated SourceRecord items: items, total, page, page_size, pages.

GET `{{API_BASE_URL}}/api/v1/admin/customers/{customer_id}/vehicles`

Local test URL: `http://localhost:8080/api/v1/admin/customers/<customer_id>/vehicles`

Authentication: Bearer admin access token

Summary: Customer Vehicles

Parameters:

- `customer_id` — path; required; string
- `page` — query; optional; integer; default: 1
- `page_size` — query; optional; integer; default: 20

Success: HTTP 200. `data` = Paginated SourceRecord items: items, total, page, page_size, pages.

GET `{{API_BASE_URL}}/api/v1/admin/garages/{garage_id}/reviews`

Local test URL: `http://localhost:8080/api/v1/admin/garages/<garage_id>/reviews`

Authentication: Bearer admin access token

Summary: Garage Reviews

Parameters:

- `garage_id` — path; required; string
- `moderation_status` — query; optional
- `page` — query; optional; integer; default: 1
- `page_size` — query; optional; integer; default: 20

Success: HTTP 200. `data` = Paginated review SourceRecord items: items, total, page, page_size, pages.

PATCH `{{API_BASE_URL}}/api/v1/admin/garages/{garage_id}/status`

Local test URL: `http://localhost:8080/api/v1/admin/garages/<garage_id>/status`

Authentication: Bearer admin access token

Summary: Garage Status

Parameters:

- `garage_id` — path; required; string

Request JSON:

```
{
  "reason": "Postman development test",
  "expected_version": 1,
  "operational_status": "ACTIVE"
}
```

Success: HTTP 200. `data` = { id, operational_status, verification_status, version, new_work_allowed }

PATCH `{{API_BASE_URL}}/api/v1/admin/garages/{garage_id}/verification`

Local test URL: `http://localhost:8080/api/v1/admin/garages/<garage_id>/verification`

Authentication: Bearer admin access token

Summary: Garage Verification

Parameters:

- `garage_id` — path; required; string

Request JSON:

```
{
  "reason": "Postman development test",
  "expected_version": 1,
  "verification_status": "UNVERIFIED"
}
```

Success: HTTP 200. `data` = { id, operational_status, verification_status, version, new_work_allowed }

GET `{{API_BASE_URL}}/api/v1/admin/jobs/delays`

Local test URL: `http://localhost:8080/api/v1/admin/jobs/delays`

Authentication: Bearer admin access token

Summary: Delays

Parameters:

- `status` — query; optional
- `overdue` — query; optional
- `page` — query; optional; integer; default: 1
- `page_size` — query; optional; integer; default: 20
- `date_from` — query; optional
- `date_to` — query; optional
- `timezone` — query; optional; string; default: UTC
- `city` — query; optional
- `garage_id` — query; optional

Success: HTTP 200. `data` = { items: [{ job_id, status, state_entered_at, state_age_hours, threshold_hours, overdue }], total, page, page_size, as_of }

GET `{{API_BASE_URL}}/api/v1/admin/notifications`

Local test URL: `http://localhost:8080/api/v1/admin/notifications`

Authentication: Bearer admin access token

Summary: Notifications

Parameters:

- `garage_id` — query; optional
- `channel` — query; optional
- `status` — query; optional
- `page` — query; optional; integer; default: 1
- `page_size` — query; optional; integer; default: 20

Success: HTTP 200. `data` = Paginated notification/delivery data: items, total, page, page_size, pages; detail returns one notification/delivery object.

GET `{{API_BASE_URL}}/api/v1/admin/notifications/{notification_id}`

Local test URL: `http://localhost:8080/api/v1/admin/notifications/<notification_id>`

Authentication: Bearer admin access token

Summary: Notification Detail

Parameters:

- `notification_id` — path; required; string

Success: HTTP 200. `data` = Paginated notification/delivery data: items, total, page, page_size, pages; detail returns one notification/delivery object.

GET `{{API_BASE_URL}}/api/v1/admin/support/tickets`

Local test URL: `http://localhost:8080/api/v1/admin/support/tickets`

Authentication: Bearer admin access token

Summary: Tickets

Parameters:

- `customer_id` — query; optional
- `status` — query; optional
- `priority` — query; optional
- `page` — query; optional; integer; default: 1
- `page_size` — query; optional; integer; default: 20
- `date_from` — query; optional
- `date_to` — query; optional
- `timezone` — query; optional; string; default: UTC
- `city` — query; optional
- `garage_id` — query; optional

Success: HTTP 200. `data` = Ticket SourceRecord or paginated ticket SourceRecord rows; mutations return updated record with incremented version.

GET `{{API_BASE_URL}}/api/v1/admin/support/tickets/{ticket_id}`

Local test URL: `http://localhost:8080/api/v1/admin/support/tickets/<ticket_id>`

Authentication: Bearer admin access token

Summary: Ticket Detail

Parameters:

- `ticket_id` — path; required; string

Success: HTTP 200. `data` = Ticket SourceRecord or paginated ticket SourceRecord rows; mutations return updated record with incremented version.

PATCH `{{API_BASE_URL}}/api/v1/admin/support/tickets/{ticket_id}/assignment`

Local test URL: `http://localhost:8080/api/v1/admin/support/tickets/<ticket_id>/assignment`

Authentication: Bearer admin access token

Summary: Ticket Assignment

Parameters:

- `ticket_id` — path; required; string

Request JSON:

```
{
  "reason": "Postman development test",
```

```
  "expected_version": 1,
  "assignee_id": "{{assignee_id}}"
}
```

Success: HTTP 200. `data` = Ticket SourceRecord or paginated ticket SourceRecord rows; mutations return updated record with incremented version.

GET `{{API_BASE_URL}}/api/v1/admin/support/tickets/{ticket_id}/messages`

Local test URL: `http://localhost:8080/api/v1/admin/support/tickets/<ticket_id>/messages`

Authentication: Bearer admin access token

Summary: Ticket Messages

Parameters:

- `ticket_id` — path; required; string
- `limit` — query; optional; integer; default: 30
- `cursor` — query; optional

Success: HTTP 200. `data` = { items: [{ id, sender_role, text, created_at, client_message_id }], next_cursor }

POST `{{API_BASE_URL}}/api/v1/admin/support/tickets/{ticket_id}/messages`

Local test URL: `http://localhost:8080/api/v1/admin/support/tickets/<ticket_id>/messages`

Authentication: Bearer admin access token

Summary: Ticket Reply

Parameters:

- `ticket_id` — path; required; string

Request JSON:

```
{
  "client_message_id": "postman-test-001",
  "text": "Development test reply. Do not send from production."
}
```

Success: HTTP 201. `data` = Message: id, sender_role, text, created_at, client_message_id (HTTP 201).

PATCH `{{API_BASE_URL}}/api/v1/admin/support/tickets/{ticket_id}/priority`

Local test URL: `http://localhost:8080/api/v1/admin/support/tickets/<ticket_id>/priority`

Authentication: Bearer admin access token

Summary: Ticket Priority

Parameters:

- `ticket_id` — path; required; string

Request JSON:

```
{
  "reason": "Postman development test",
  "expected_version": 1,
  "priority": "NORMAL"
}
```

Success: HTTP 200. `data` = Ticket SourceRecord or paginated ticket SourceRecord rows; mutations return updated record with incremented version.

PATCH `{{API_BASE_URL}}/api/v1/admin/support/tickets/{ticket_id}/status`

Local test URL: `http://localhost:8080/api/v1/admin/support/tickets/<ticket_id>/status`

Authentication: Bearer admin access token

Summary: Ticket Status

Parameters:

- `ticket_id` — path; required; string

Request JSON:

```
{
  "expected_version": 1,
  "status": "IN_PROGRESS"
}
```

Success: HTTP 200. `data` = Ticket SourceRecord or paginated ticket SourceRecord rows; mutations return updated record with incremented version.

Admin Owner Statistics

GET `{{API_BASE_URL}}/api/v1/admin/garage-owner-statistics`

Local test URL: `http://localhost:8080/api/v1/admin/garage-owner-statistics`

Authentication: Bearer admin access token

Summary: Owner Statistics

Parameters:

- `snapshot_id` — query; optional
- `exclude_inactive` — query; optional; boolean; default: false
- `sort` — query; optional; string; allowed values: jobs_created_desc, net_collected_amount_desc, owner_name_asc; default: jobs_created_desc
- `page` — query; optional; integer; default: 1
- `page_size` — query; optional; integer; default: 20
- `date_from` — query; optional
- `date_to` — query; optional
- `timezone` — query; optional; string; default: UTC
- `city` — query; optional
- `garage_id` — query; optional

Success: HTTP 200. `data` = Snapshot metadata and owner metric row(s); list includes items/total/page/page_size/source_totals; trend returns series[]. Finance metrics require finance.read.

GET `{{API_BASE_URL}}/api/v1/admin/garage-owner-statistics/summary`

Local test URL: `http://localhost:8080/api/v1/admin/garage-owner-statistics/summary`

Authentication: Bearer admin access token

Summary: Owner Summary

Parameters:

- `snapshot_id` — query; optional
- `exclude_inactive` — query; optional; boolean; default: false
- `date_from` — query; optional

- `date_to` — query; optional
- `timezone` — query; optional; string; default: UTC
- `city` — query; optional
- `garage_id` — query; optional

Success: HTTP 200. `data` = Snapshot metadata and owner metric row(s); list includes items/total/page/page_size/source_totals; trend returns series[]. Finance metrics require finance.read.

GET `{{API_BASE_URL}}/api/v1/admin/garage-owner-statistics/{owner_id}`

Local test URL: `http://localhost:8080/api/v1/admin/garage-owner-statistics/<owner_id>`

Authentication: Bearer admin access token

Summary: Owner Detail

Parameters:

- `owner_id` — path; required; string
- `snapshot_id` — query; optional
- `exclude_inactive` — query; optional; boolean; default: false
- `date_from` — query; optional
- `date_to` — query; optional
- `timezone` — query; optional; string; default: UTC
- `city` — query; optional
- `garage_id` — query; optional

Success: HTTP 200. `data` = Snapshot metadata and owner metric row(s); list includes items/total/page/page_size/source_totals; trend returns series[]. Finance metrics require finance.read.

GET `{{API_BASE_URL}}/api/v1/admin/garage-owner-statistics/{owner_id}/trends`

Local test URL: `http://localhost:8080/api/v1/admin/garage-owner-statistics/<owner_id>/trends`

Authentication: Bearer admin access token

Summary: Owner Trends

Parameters:

- `owner_id` — path; required; string
- `snapshot_id` — query; optional
- `exclude_inactive` — query; optional; boolean; default: false
- `period` — query; optional; string; allowed values: daily, weekly, monthly; default: daily
- `date_from` — query; optional
- `date_to` — query; optional
- `timezone` — query; optional; string; default: UTC
- `city` — query; optional
- `garage_id` — query; optional

Success: HTTP 200. `data` = Snapshot metadata and owner metric row(s); list includes items/total/page/page_size/source_totals; trend returns series[]. Finance metrics require finance.read.

Admin Reports

POST `{{API_BASE_URL}}/api/v1/admin/reports/exports`

Local test URL: `http://localhost:8080/api/v1/admin/reports/exports`

Authentication: Bearer admin access token

Summary: Request Export

Parameters:

- `Idempotency-Key` — header; required; string

Request JSON:

```
{
  "report_type": "OPERATIONS",
  "format": "CSV",
  "filters": {
    "date_from": "2026-10-01",
    "date_to": "2026-10-13",
    "timezone": "UTC"
  },
  "exclude_inactive": false
}
```

Success: HTTP 202. `data` = Export job: id, status, report_type, format, filters, snapshot_id, row_count, source_totals, metric_definitions, expires_at, error, download_path.

GET `{{API_BASE_URL}}/api/v1/admin/reports/exports/{export_id}`

Local test URL: `http://localhost:8080/api/v1/admin/reports/exports/<export_id>`

Authentication: Bearer admin access token

Summary: Report Status

Parameters:

- `export_id` — path; required; string

Success: HTTP 200. `data` = Export job: id, status, report_type, format, filters, snapshot_id, row_count, source_totals, metric_definitions, expires_at, error, download_path.

GET `{{API_BASE_URL}}/api/v1/admin/reports/exports/{export_id}/download`

Local test URL: `http://localhost:8080/api/v1/admin/reports/exports/<export_id>/download`

Authentication: Bearer admin access token

Summary: Download Report

Parameters:

- `export_id` — path; required; string

Success: HTTP 200. `data` = CSV/XLSX attachment bytes with Content-Disposition; not JSON.

Admin Security

GET `{{API_BASE_URL}}/api/v1/admin/permissions`

Local test URL: `http://localhost:8080/api/v1/admin/permissions`

Authentication: Bearer admin access token

Summary: Permissions

Success: HTTP 200. `data` = { items: [{ code, description }] }

GET `{{API_BASE_URL}}/api/v1/admin/roles`

Local test URL: `http://localhost:8080/api/v1/admin/roles`

Authentication: Bearer admin access token

Summary: Roles

Success: HTTP 200. `data` = { items: [{ id, name, permissions: string[] }] }

GET `{{API_BASE_URL}}/api/v1/admin/staff`

Local test URL: `http://localhost:8080/api/v1/admin/staff`

Authentication: Bearer admin access token

Summary: Staff

Parameters:

- `search` — query; optional
- `is_active` — query; optional
- `page` — query; optional; integer; default: 1
- `page_size` — query; optional; integer; default: 20

Success: HTTP 200. `data` = { items: AdminStaff[], total, page, page_size, pages }

PUT `{{API_BASE_URL}}/api/v1/admin/staff/{admin_id}/roles`

Local test URL: `http://localhost:8080/api/v1/admin/staff/<admin_id>/roles`

Authentication: Bearer admin access token

Summary: Assign Roles

Parameters:

- `admin_id` — path; required; string

Request JSON:

```
{
  "role_ids": [
    "role_operations"
  ],
  "data_scope": {
    "type": "PLATFORM",
    "garage_ids": []
  },
  "reason": "Postman development test; assign only to a separate test admin."
}
```

Success: HTTP 200. `data` = Updated AdminStaff object: { id, display_name, email, is_active, role_ids, permissions, data_scope, mfa_enrolled, mfa_required }

Frontend integration notes

- Use the `version` returned by a GET before garage or ticket changes; never assume version 1. Stale updates return 409.
- Role IDs are `role_ceo`, `role_operations`, `role_finance`, `role_support`. Finance routes/fields require `finance.read`.
- New financial metrics use integer INR paise (■1 = 100 paise); existing dashboard/job revenue preserves recorded-job rupees.

- Real customer, payment/refund, review, and delivery rows require authentic source data. Empty data is not proof of no upstream transactions.
- Announcement sends currently support IN_APP only; PUSH/SMS/EMAIL return 409. Send is a real inbox mutation and requires `Idempotency-Key`.
- Exports are asynchronous: request -> poll status -> download when READY.
- This is the development API contract. Do not send mutation examples to production without separate approval.